PAPER_561: Buoyancy-Stratified Factorial Geometry — Black Hole Horizon Solution

Unified Quantum Field Framework (UQFF)

Daniel T. Murphy • UQFF Research Consortium • 2026

Source: PAPER_561_BSFG_BlackHole_Horizon_Solution.md

PAPER_561: Buoyancy-Stratified Factorial Geometry — Black Hole Horizon Solution

---

Author: Daniel T. Murphy — Star Magic / UQFF Framework **Session:** 149 | **Source:** Composed from CP4 #149, #147 (Sessions 148, 147) **CP4 Class:** `BSFGBlackHoleSolutionHorizonCalculator` (#156) **Date:** 2026-03-27

> ***Context note:*** *The BSFG metric $A_{\mu\nu}(r)$ from CP4 #149 has a time-like component $A_{00}(r) = 1 + \eta T_{s00}(r)\cos(\pi t_n)$. A metric horizon occurs where $g_{tt} = A_{00}(r_h) = 0$. This paper solves this condition analytically, derives the BSFG surface gravity and Hawking temperature, and contrasts the result with the proplyd equilibrium radius r_q from PAPER_550 (CP4 #147).*

---

§1 Abstract

We solve the BSFG horizon equation $A_{00}(r_h) = 0$ and derive:

$$r_h = \left(\eta C_{\rm num}|\cos(\pi t_n)|\right)^{1/3}, \qquad \text{(physical when } \cos(\pi t_n) < 0\text{)}$$

At $t_n = 1$ (maximum Aether anti-phase): $r_h \approx 1.62 \times 10^8\ {\rm m} \approx 0.233\,R_\odot$. The BSFG surface gravity and Hawking temperature are:

$$\kappa_{\rm BSFG} = \frac{3c^2\eta|C_{\rm num}||\cos(\pi t_n)|}{2r_h^4} = \frac{3c^2}{2r_h}, \qquad T_H^{\rm BSFG} = \frac{\hbar\kappa_{\rm BSFG}}{2\pi k_B c} \approx 3.37 \times 10^{-12}\ {\rm K}$$

The scale hierarchy: $r_h \approx 0.23\,R_\odot \ll r_q \approx 0.097\ {\rm AU}$ — the BSFG horizon (when it exists) lies deep inside the star, ~145× smaller than the proplyd equilibrium radius.

§2 Horizon Condition

The BSFG metric time-time component (CP4 #149):

$$A_{00}(r) = 1 + \varepsilon(r) = 1 + \frac{\eta C_{\rm num}\cos(\pi t_n)}{r^3}$$

Setting $A_{00}(r_h) = 0$:

$$1 + \frac{\eta C_{\rm num}\cos(\pi t_n)}{r_h^3} = 0 \implies r_h^3 = -\eta C_{\rm num}\cos(\pi t_n)$$

Physical requirement: $r_h^3 > 0$ demands $\cos(\pi t_n) < 0$, i.e.:

$$\frac{1}{2} < t_n < \frac{3}{2} \pmod{2} \quad \text{(Aether anti-phase)}$$

In the $t_n \in [0,2)$ Aether cycle, a horizon only exists during the anti-phase half. During the normal phase $(\cos > 0)$, the Aether is repulsive and no horizon forms.

§3 Horizon Radius

Step 1. At $t_n = 1$ ($\cos(\pi t_n) = -1$):

$$r_h = (\eta C_{\rm num})^{1/3}$$

Substituting $\eta = 10^{-22}\ {\rm m^3/J}$ and $C_{\rm num} \approx 4.27 \times 10^{46}\ {\rm J}$:

$$r_h = (10^{-22} \times 4.27 \times 10^{46})^{1/3} = (4.27 \times 10^{24})^{1/3} \approx 1.62 \times 10^8\ {\rm m}$$

Step 2. Scale comparisons:

Length scale	Value	Ratio to r_h
r_h (BSFG horizon)	1.62×10^8 m	1
$R_\odot$ (solar radius)	6.96×10^8 m	$\times 4.3$
r_q (proplyd equilibrium)	1.45×10^{10} m	$\times 90$
$R_{s,\rm GR}$ (Schwarzschild)	2.95×10^3 m	$\times 5.5 \times 10^{-5}$

The BSFG horizon is ~55,000 times larger than the GR Schwarzschild radius — but lies inside the stellar interior (0.233 $R_\odot$), so it is only relevant for compact objects.

§4 Surface Gravity and Hawking Temperature

Step 3. BSFG surface gravity (from metric derivative at r_h):

$$\kappa_{\rm BSFG} = \frac{c^2}{2}\left|\frac{\partial A_{00}}{\partial r}\right|_{r_h} = \frac{c^2}{2} \cdot \frac{3\eta|C_{\rm num}||\cos(\pi t_n)|}{r_h^4}$$

Using $r_h^3 = \eta|C_{\rm num}||\cos|$:

$$\kappa_{\rm BSFG} = \frac{3c^2}{2r_h} \approx \frac{3 \times (3 \times 10^8)^2}{2 \times 1.62 \times 10^8} \approx 8.33 \times 10^8\ {\rm m\,s}^{-2}$$

Step 4. BSFG Hawking temperature:

$$T_H^{\rm BSFG} = \frac{\hbar\kappa_{\rm BSFG}}{2\pi k_B c} = \frac{1.055 \times 10^{-34} \times 8.33 \times 10^8}{2\pi \times 1.381 \times 10^{-23} \times 3 \times 10^8} \approx 3.37 \times 10^{-12}\ {\rm K}$$

For comparison, the GR Hawking temperature for a solar-mass black hole:

$$T_H^{\rm GR}(M_\odot) = \frac{\hbar c^3}{8\pi G M_\odot k_B} \approx 6.17 \times 10^{-8}\ {\rm K}$$

The BSFG Hawking temperature is ~18,000 times colder than the GR result — consistent with a much larger horizon radius.

---

§5 Physical Interpretation

1. **The BSFG horizon is phase-dependent.** It only exists during the Aether anti-phase $(\cos(\pi t_n) < 0)$, appearing and disappearing on the Aether oscillation timescale. This "blinking horizon" has no GR analog.
2. **The horizon lies inside stellar matter.** For a solar-type star, $r_h \approx 0.23\,R_\odot$. The BSFG horizon is only physically accessible for compact objects where the stellar radius $r_* < r_h$, requiring a density $\rho_* > 3M_\odot/(4\pi r_h^3) \approx 5.6 \times 10^5\ {\rm kg\,m}^{-3}$ — white dwarf density range.
3. **Distinct from r_q.** The proplyd equilibrium radius $r_q \approx 0.097\ {\rm AU}$ (PAPER_550) is where $U_m = 0$ — a force equilibrium in the DPM field. The BSFG horizon is where the metric degeneracy condition $A_{00} = 0$ is met — a purely geometric criterion. The two coincide only by fine-tuning of Aether parameters.

---

§6 References

- CP4 #149 — `BSFGRiemannCurvatureAetherMetricCalculator` — PAPER_554 ($A_{00}(r)$ metric)
- CP4 #147 — `Um26DPolyQuantizationDPMConfinementCalculator` — PAPER_550 ($r_q = 0.097$ AU)
- CP4 #150 — `BSFGGeodesicMetricCompatibilityCalculator` — PAPER_555 (geodesic equation)

- `bh_thermodynamics_module.py` — Hawking temperature framework (GR comparison)